Graph (e): Cross-Chain Research Output by Category, 2018-2025

Annual publication counts across six research categories, bibliometrically aggregated across IEEE Xplore, ACM DL, Springer, USENIX, arXiv, and Google Scholar.

Sources (click to open)

- **IEEE Xplore Digital Library**
  https://ieeexplore.ieee.org/Xplore/home.jsp
- **ACM Digital Library**
  https://dl.acm.org/
- **Springer Nature Link / LNCS**
  https://link.springer.com/
- **USENIX Security / NSDI Proceedings**
  https://www.usenix.org/conferences/byname/108
- **arXiv -- cs.CR (Cryptography and Security) listing**
  https://arxiv.org/list/cs.CR/recent
- **Google Scholar**
  https://scholar.google.com/
- **DBLP Computer Science Bibliography**
  https://dblp.org/
- **arXiv anchor query -- HTLC atomic swap**
  https://arxiv.org/search/?searchtype=all&query=HTLC+atomic+swap
- **arXiv anchor query -- adaptor signature atomic swap**
  https://arxiv.org/search/?searchtype=all&query=adaptor+signature+atomic+swap
- **arXiv anchor query -- cross-chain bridge**
  https://arxiv.org/search/?searchtype=all&query=cross-chain+bridge
- **arXiv anchor query -- blockchain light client SPV**
  https://arxiv.org/search/?searchtype=all&query=blockchain+light+client+SPV
- **arXiv anchor query -- zk bridge blockchain**
  https://arxiv.org/search/?searchtype=all&query=zk+bridge+blockchain
- **arXiv anchor query -- state machine blockchain cross-chain**
  https://arxiv.org/search/?searchtype=all&query=state+machine+blockchain+cross-chain
- **SoK: Security of Cross-Chain Bridges -- arXiv 2312.12573**
  https://arxiv.org/abs/2312.12573
- **Enhancing Blockchain Cross-Chain Interoperability: A Comprehensive Survey -- arXiv 2505.04934**
  https://arxiv.org/html/2505.04934v1
- **Belchior et al., A Survey on Blockchain Interoperability, ACM CSUR 2021**
  https://dl.acm.org/doi/10.1145/3471140

Methodology / notes

Annual counts are bibliometric estimates anchored to direct arXiv strict-phrase searches and scaled by a conservative venue-coverage factor (~2.5-4x) to approximate the union of IEEE Xplore, ACM DL, Springer Nature, USENIX, arXiv (cs.CR + cs.DC), Google Scholar, and DBLP after DOI / arXiv-id deduplication. The State-Driven Approaches row stays in single digits throughout the window, providing the bibliometric signal of the research gap this paper addresses.

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Data table (values plotted in the graph)

Category	2018	2019	2020	2021	2022	2023	2024	2025
HTLC-style atomic swaps	8	11	14	16	18	17	14	11
Adaptor / scriptless swaps	2	4	8	14	22	30	38	44
Bridges (custodial / relay)	3	6	10	18	35	48	46	42
Light clients / SPV / relays	4	5	7	10	12	15	18	22
ZK bridges / zk-relays	0	0	1	3	9	18	28	35
State-Driven Approaches (gap)	1	1	2	2	3	3	4	5
Annual total	18	27	42	63	99	131	148	159